

Experimental Section

First, the oscilloscope's channel 2 was connected to the laser module's sync jack to identify the trigger signal. The horizontal scale was adjusted to 1 μ s per grid, the voltage scale was autoranged, and was set so the oscilloscope triggers on rising laser module sync voltage so the measurement happens after the laser turns off. Then, the trigger frequency was recorded to be about 1.044 kHz.

Then, the laser and sample holder was set up (as described in diagram below), first to check that we can see the laser turn on and off through scattered laser light. A piece of tape was put on an empty sample tube to do this, and the photodiode detector signal was connected to ch1 of the oscilloscope. A square signal was successfully observed with this method.

The following solutions were mixed using micro-pipettes and stock solutions of 2.5mM Ru(bpy)₃, 1.25 methyl viologen dichloride (quencher), and 0.5M of sulfuric acid:

Sample	Volume of Ru(bpy) ₃ (uL)	Volume of Quencher (uL)	Volume of Acid (uL)	Total Volume (uL)	[Ru(bpy) ₃] (mM)	[Methyl Viologen] (mM)	H ₂ SO ₄ (M)
1	700	0	250	1000	1.75	0	0.125
2	700	50	250	1000	1.75	0.625	0.125
3	700	100	200	1000	1.75	1.25	0.1
4	700	150	150	1000	1.75	1.875	0.075
5	700	200	100	1000	1.75	2.5	0.05

Table 1: volumes of stock solutions needed to create each solution with varying quencher concentrations, along with their total concentrations

Each sample was put in a NMR tube and placed into the laser setup, slightly realigning the angle of the detector such that a "shark fin" pattern is visible. There were also vertical rises and drops due to the reflected laser light entering the detector instead of phosphorescent light, which had to be accounted for by adjusting the detector position.

Diagram of Laser Setup

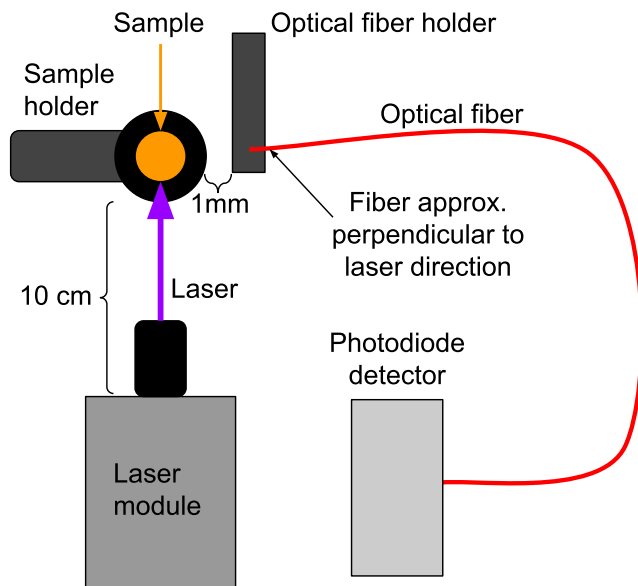


Fig 1. Diagram of laser and photodiode detector setup and their relative positions and orientation to the sample solution under test

Data Analysis

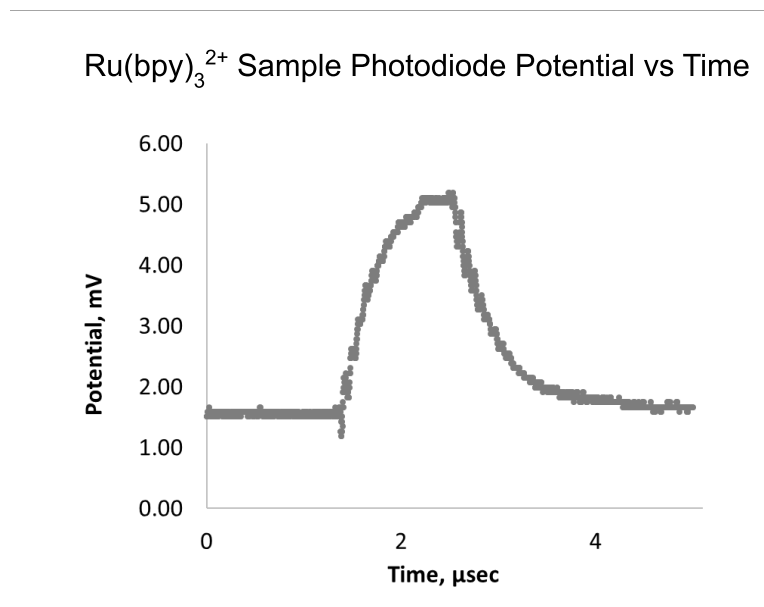


Fig 2. Phosphorescence over time of a $\text{Ru}(\text{bpy})_3^{2+}$ sample with no quencher exposed to a short burst of laser, where the voltage and time values have been shifted for data analysis purposes such that the minimum voltage is above 1mV and time is positive

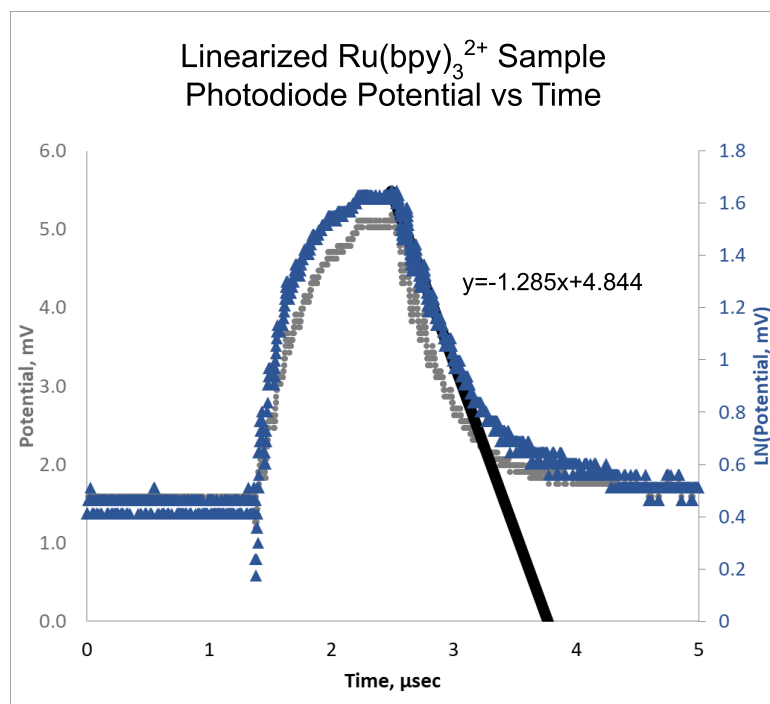


Fig 3. Phosphorescence over time of a $\text{Ru}(\text{bpy})_3^{2+}$ sample with no quencher, where data from Fig 2. has been transformed with the log function and the originally exponentially decaying segment (now linear) is fit to a line. The negative of the slope is the k_{obs} , the observed decay rate, which is 1.285 ± 0.015

Linearized Phosphorescence over Time of $\text{Ru}(\text{bpy})_3^{2+}$
Samples with Varying Quencher Concentrations

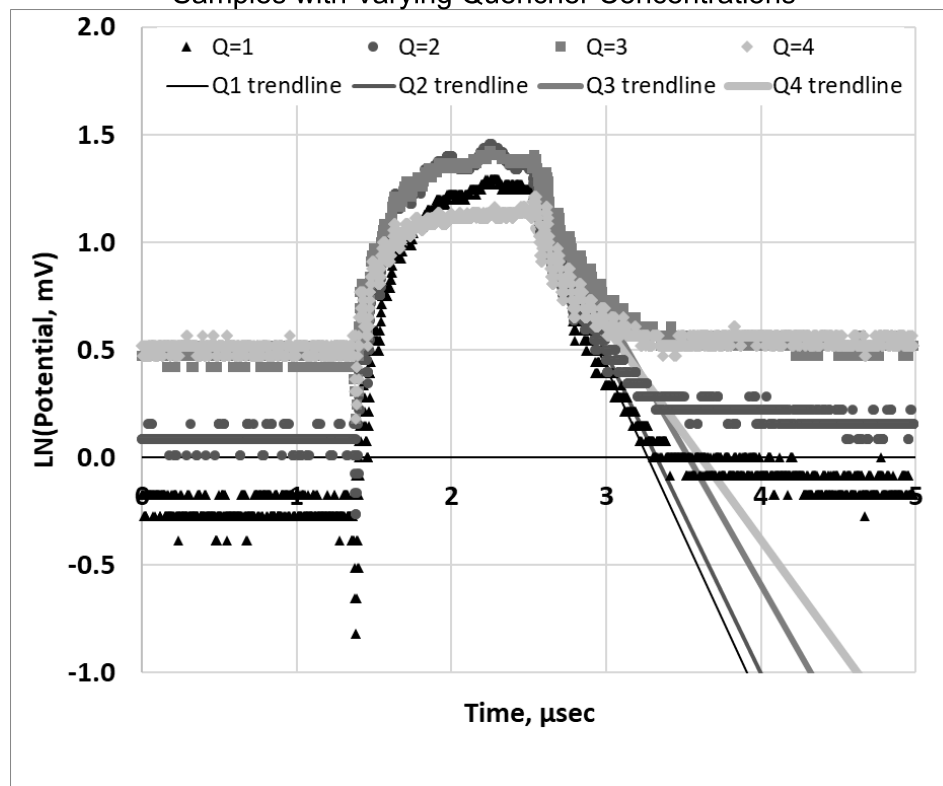


Fig 4. Other solutions' phosphorescence over time where Q1 has 0.625mM of quencher, Q2 has 0.125 mM quencher, Q3 has 0.250 mM quencher, and Q4 has 0.500 mM quencher, along with each of their linear segments fit to a line. Q1's slope is -1.285, Q2's is -1.561, Q3's is -1.490, Q4's is -0.987

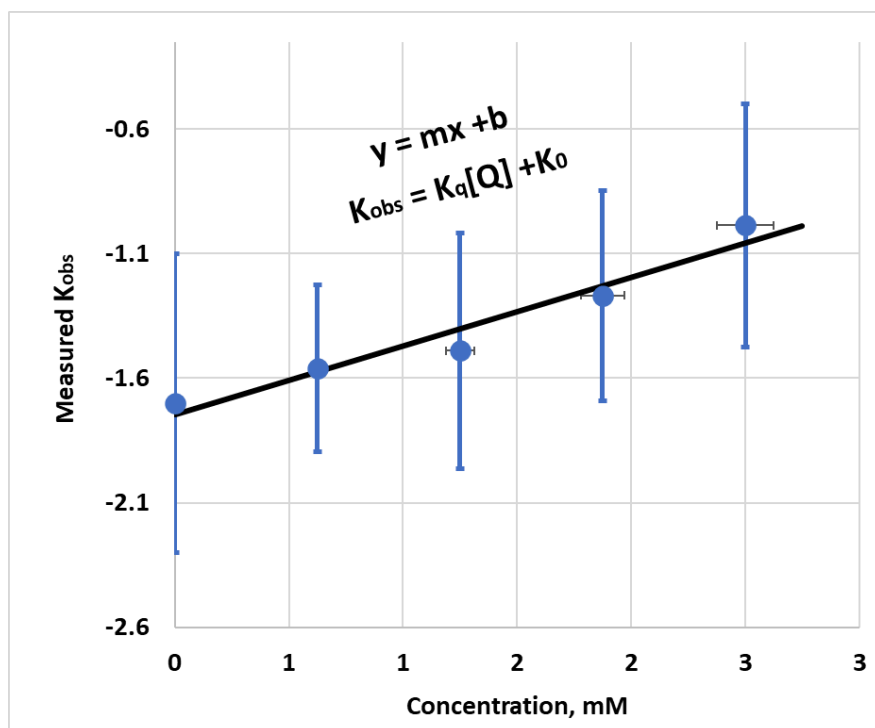


Fig 5. K_{obs} vs different quencher concentrations, where k_q is 0.142 here